

dnf & rpm Cheat Sheet

Install, update and undo software changes on RHEL, Rocky, Alma and Fedora.

Using it

```
dnf check-update    ← lists available updates; installs nothing
sudo dnf upgrade    ← installs all updates (refreshes the repo data itself)
sudo dnf upgrade --security ← security fixes only (RHEL and clones)
```

Command	Does	Command	Does
dnf install pkg	Installs it + its deps	dnf search word	Searches all packages
dnf install ./app.rpm	Local file + its deps	dnf info pkg	Version, size, repo
dnf remove pkg	Uninstalls it	dnf provides '*/bin/dig'	Which package has it
dnf autoremove	Drops unused deps	dnf list installed	Everything installed
dnf reinstall pkg	Restores its files	dnf list available	What you could install
dnf downgrade pkg	Older version	dnf repolist	Enabled repos
dnf group install "Server"	Installs a set	dnf clean all	Clears the cache
dnf versionlock add pkg	Freezes its version*	dnf makecache	Rebuilds the cache

* dnf4 needs **python3-dnf-plugin-versionlock** (dnf5: built in). **versionlock delete pkg** frees it.

Reading dnf history (RHEL 9 / dnf4 layout)

ID	Command line	Date and time	Action(s)	Altered
42	install nginx	2025-09-30 14:02	Install	4 ← 4 packages
41	upgrade	2025-09-28 09:15	I, U	37 ← I = installed, U = upgraded
40	remove httpd	2025-09-20 11:40	Removed	6

dnf history info 42 = details · **sudo dnf history undo 42** = reverses it · **rollback 40** = back to after 40

Querying with rpm

Command	Shows	Command	Shows
rpm -qa	All installed packages	rpm -qf /usr/bin/dig	Owner: bind-utils
rpm -qi pkg	Info: version, build date	rpm -qc openssh-server	Its config files
rpm -ql pkg	Every file it installed	rpm -qa --last head	Newest installs first
rpm -V pkg	Files changed since install	rpm --import KEY.asc	Trusts a repo signing key

Reading sudo rpm -Va (verify; no output = nothing changed)

```
S.5....T.  c /etc/ssh/sshd_config ← you edited a config file (c): normal
.M.....  /var/log/httpd        ← permissions changed
..5....T.  /usr/bin/ls           ← a program changed: investigate
```

S size · **M** mode (permissions) · **5** checksum · **D** device · **L** link · **U** user · **G** group · **T** time · **P** caps · **.** = unchanged

Repos and EPEL

Command	Does
sudo dnf config-manager --set-enabled crb	Enables CRB, needed by EPEL (Rocky/Alma 9)
sudo dnf install epel-release	Adds EPEL (on RHEL itself: see EPEL's docs)
sudo dnf config-manager setopt crb.enabled=1	Same as line 1, in dnf5 (Fedora 41+)

Repo files: **/etc/yum.repos.d/*.repo** · dnf5 also renames: **list --installed, config-manager addrepo**

Handy recipes

Command	Does
sudo dnf needs-restarting -r	Is a reboot needed? (RHEL 9 / dnf4)
sudo systemctl enable --now dnf-automatic-install.timer	Auto-installs updates (RHEL 9)

Remember: add only repos you trust. EPEL is the standard one for RHEL clones; others may replace system packages.